100 companies • 92 sector-classified • 10 sectors • 2011–2024 financials • Streamlit + Valuation + Screener

FINANCIAL INTELLIGENCE PLATFORM

Analyst Guide

Comprehensive instructions for using the 8-screen Streamlit dashboard, screener, peer comparison, valuation and reporting workflows — covering 100 canonical companies, of which 92 are sector-classified across 10 sectors (2011–2024).

Version	v4.0
Coverage	100 tickers • 92 sector-classified • 10 sectors
Warehouse	1,263 P&L; / 1,137 BS / 1,152 CF / 5,520 prices
Stack	Streamlit 1.39 • Plotly 5.24 • SQLite • ReportLab • FastAPI
Run	streamlit run src/dashboard/app.py --server.port 8501

Data through FY2024 • Monetary values in INR Crore • SIMULATED datasets labelled where applicable.

Dashboard: http://localhost:8501 | API: uvicorn src.api.main:app --port 8000

Prepared for analysts, portfolio managers and investment researchers.

All charts use use_container_width=True and handle missing data as N/A — never crash on NaN.

100 companies • 92 sector-classified • 10 sectors • 2011–2024 financials • Streamlit + Valuation + Screener

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How to use this guide: Chapters 1–3 are read-once onboarding. Chapters 4–10 are task-oriented — jump directly to the screen you need. Chapter 11 is a reference for ratio definitions and valuation flags.

Conventions: bold = UI element | italic = file/path | code = command | N/A = missing data (graceful handling).

CHAPTER 1

Introduction & Platform Overview

Purpose. The Financial Intelligence Platform is a financial-intelligence workbench that consolidates 14 years (2011–2024) of audited financials for a universe of 100 companies, of which 92 are sector-classified across 10 GICS-aligned sectors. It lets an analyst move from market-wide screening to single-company deep dives without leaving the browser — all 8 screens share the same SQLite warehouse (data/nifty100.db) and the same cached data layer.

The platform tracks 100 canonical tickers in the companies table; 92 of them carry a broad_sector / sub_sector classification that drives every downstream sector chart, while the remaining 8 are retained for reference only. Sector weights are stored in companies.broad_sector / sub_sector.

Current sector distribution of the 92 classified companies: Financials 23, Energy 14, Consumer Discretionary 14, Industrials 10, Materials 9, Consumer Staples 7, Healthcare 6, Information Technology 5, Real Estate 2, Communication Services 2.

Who Should Use It

- **Equity analysts** preparing initiation or quarterly update notes — use Company Profile for the KPI snapshot and Peer Comparison radar to benchmark margins and leverage.
- **Portfolio managers & screeners** hunting for Quality / Value / Dividend ideas — Screener with 10 sliders + 6 presets and CSV export maps directly to an order-management system.
- **Risk & credit teams** checking leverage and coverage — Sector Analysis bubble (Revenue vs ROE, size = Market Cap) and Capital Allocation Map treemap surface outlier leverage in one glance.
- **Retail / wealth researchers** producing client tearsheets — one click generates a 2-page PDF per ticker (reports/tearsheets/{ticker}_tearsheet.pdf).

Key Capabilities at a Glance

8-screen Streamlit dashboard: Home, Profile, Screener, Peers, Trends, Sectors, Capital Map, Reports — all wide layout, sidebar expanded.

10-year financial lens: Profit & Loss, Balance Sheet, Cash Flow and computed ratios per ticker-year; YoY % annotations and CAGR 5Y on every trend.

Screener with presets: 10 metric sliders (ROE min, D/E max, FCF min, Rev CAGR min, PAT CAGR min, OPM min, P/E max, P/B max, Div Yield min, ICR min) + Quality / Value / Growth / Dividend / Debt-Free / Turnaround one-click presets.

Peer & sector benchmarking: 11 peer groups (56 memberships) with fallback to broad_sector; radar normalized 0–1 with D/E inverted; sector-median KPI bars.

Capital discipline view: 8 allocation patterns derived from CFO/CFI/CFF signs (Healthy Growth, Aggressive Growth, Mature Returner, etc.), treemap sized by |CFO|, coloured by sector.

Valuation flags: FCF yield, sector-median P/E, automatic Caution / Discount / Fair flag (output/valuation_summary.xlsx, 100 rows).

Design Principles

1. Graceful degradation — some companies have <10 years of ratios. Charts show available history with an info banner; any NaN/None renders as N/A rather than crashing.

2. Speed — every loader in src/dashboard/utils/db.py is decorated with @st.cache_data(ttl=600); a Company Profile load is well under the 3s SLA.

3. Reproducibility — ETL is prefix-agnostic, header-row auto-detecting, and FK-checked. make load; make ratios; make test reproduces the warehouse from the Excel sources in data/source/.

Quick orientation: If you have 5 minutes, open Home → switch the sidebar year → search a ticker in Company Profile → click Quality in Screener → pick IT Services in Peers. You will have touched the four most-used paths.

What This Guide Covers

Chapter	You will learn to
2	Understand where the numbers come from and how they are cleaned
3	Install, run and navigate the 8 screens

Chapter	You will learn to
4–6	Monitor the market, drill into a company, and screen for ideas
7–9	Benchmark vs peers/sectors and diagnose capital allocation
10–11	Pull annual reports, generate tearsheets and interpret valuation flags
A	Cross-reference every deliverable against live evidence

CHAPTER 2

Data Architecture & Sources

The entire platform is file-driven. Place the Excel sources in data/source/ and run python -m src.etl.loader (or make load). The loader discovers files by suffix (*companies.xlsx, *balancesheet.xlsx, …) so any hash prefix is ignored.

Source Files

Source file	Table	Rows	Notes
companies.xlsx	companies	100	company_id = ticker; broad_sector / sub_sector
profitandloss.xlsx	profitandloss	1,263	Company-year P&L; normalize_year applied
balancesheet.xlsx	balancesheet	1,137	CWIP + other_asset merged into other_assets
cashflow.xlsx	cashflow	1,152	CFO / CFI / CFF signs drive Capital Map
financial_ratios.xlsx	financial_ratios	1,263	45 columns; recomputed by make ratios
market_cap.xlsx	stock_prices	5,520	Real + synthetic rows; SIMULATED
documents.xlsx	documents	1,584	BSE annual-report URLs (report_url)
analysis.xlsx	analysis	0	CAGR text columns
sectors.xlsx	sectors	10	Sector lookup
prosandcons.xlsx	prosandcons	26	Pros / cons per company
peer_groups.xlsx	peer_groups	56	Group membership + benchmark flag

ETL Pipeline — How It Works

1. **read_excel_smart(path)** tries header=1, then header=0, detecting numeric/Unnamed columns and falling back automatically.
2. **Normalization** (src/etl/normaliser.py): normalize_year handles Mar-13→2013, 2020-21→2020, Dec 2012→2012; normalize_ticker trims/uppercases; column mapping merges cwip into other_assets and maps other_asset / Annual_Report / about_company to their canonical names.
3. **Two-phase load:** Phase 1 loads real tables; Phase 2 fills missing tables only.
4. **FK integrity:** child tables are filtered to parent companies; PRAGMA foreign_key_check = 0; load_audit.csv and validation_failures.csv are written to output/.

SQLite Schema (data/nifty100.db)

Table	Key columns	Grain
companies	company_id (PK ticker), company_name, broad_sector, sub_sector, about	One row per company
profitandloss	company_id, year, sales, net_profit + full P&L;	Company-year
balancesheet	company_id, year, total_assets, debt, equity, other_assets	Company-year
cashflow	company_id, year, cfo, cfi, cff, free_cash_flow_cr	Company-year
financial_ratios	company_id, year, roe, roce, npm, de, pe, pb, icr, cagr cols	Company-year (computed)
stock_prices	company_id, date, close_price, market_cap_crore, pe_ratio, ev_ebitda	Company-date
documents	company_id, year, report_url	Company-year (BSE PDFs)
peer_groups	peer_group_name, company_id, is_benchmark	Group membership
sectors / analysis	sector lookup + CAGR text cols	Reference

Monetary unit: All values are stored in INR Crore. **SIMULATED label:** the stock_prices / market_cap datasets are tagged SIMULATED in the warehouse — treat valuation outputs as indicative, not live market data.

Data-quality note: output/validation_failures.csv contains 329 row(s) (0 CRITICAL). Warnings are non-blocking; the loader proceeds and records them for review.

CHAPTER 3

Getting Started — Installation & Navigation

Install once, run everywhere — setup, make targets and the 8-screen map.

Prerequisites: Python 3.11+, 1 GB free disk, no Docker required. Dependencies are pinned in requirements.txt (pandas, streamlit, plotly, reportlab, fastapi).

Setup (Windows / macOS / Linux)

Step	Command	What it does
1. Clone / unzip	—	Place *.xlsx sources in data/source/
2. Virtualenv	python -m venv venv (venv\Scripts\activate on Windows)	Isolated environment
3. Install	pip install -r requirements.txt	Installs all pinned deps
4. Load DB	make load — or — python -m src.etl.loader	Builds data/nifty100.db (100 companies)
5. Ratios	make ratios	Computes financial_ratios (1,263 rows)
6. Test	pytest tests/ -v	147 tests; sanity-checks ETL & KPIs
7. Dashboard	streamlit run src/dashboard/app.py --server.port 8501	Opens http://localhost:8501
8. API (optional)	uvicorn src.api.main:app --port 8000	REST API at http://localhost:8000

If data/nifty100.db is missing the landing page shows a red banner: “Database not found — run make load first.”

Project Layout

- *src/dashboard/app.py* — entry point, page_config wide / title / sidebar expanded.
- *src/dashboard/utils/db.py* — cached loaders: get_companies, get_ratios, get_pl, get_bs, get_cf, get_sectors, get_peers, get_valuation.
- *src/dashboard/pages/01_home.py … 08_reports.py* — the 8 screens.
- *src/analytics/valuation.py* — FCF yield + sector P/E flags.
- *output/valuation_summary.xlsx* & *output/valuation_flags.csv*.
- *config/screener_config.yaml* — preset thresholds.
- *reports/tearsheets/{ticker}_tearsheet.pdf* — 2-page per-company PDFs.

The 8-Screen Map

#	Screen	File	Use when you want to…
1	Home	01_home.py	Get a market snapshot for a chosen year
2	Company Profile	02_profile.py	Deep-dive a single ticker (10-yr trends)
3	Screener	03_screener.py	Filter the universe by 10 financial thresholds
4	Peer Comparison	04_peers.py	Benchmark a company vs its sector peers
5	Trend Analysis	05_trends.py	Overlay up to 3 metrics and see YoY %
6	Sector Analysis	06_sectors.py	Compare sectors via bubble & median bars
7	Capital Allocation Map	07_capital.py	See how cash is sourced & deployed
8	Annual Reports	08_reports.py	Pull BSE PDFs and tearsheets

Navigation Conventions

- **Sidebar is always visible** (initial_sidebar_state=expanded) and holds the primary controls: year selector on Home, 10 sliders on Screener, search boxes elsewhere.
- **Two navigation paths:** sidebar radio and quick-nav buttons on the landing page — both call st.switch_page with a graceful fallback.
- **Charts never overflow** — every Plotly figure uses use_container_width=True.
- **Missing data = N/A** — never a traceback. If a ticker has <10 years, an info banner states the available history and the chart shows only available points.

CHAPTER 4

Home — Market Overview

Open Home (pages/01_home.py). At the top you see a subtitle “Market Overview” and a sidebar control labelled **Select Year** (default the latest year). Changing the year recomputes every KPI and the Top-5 table.

The KPI Tiles

Tile	Definition	How to read it
Average ROE	Mean return_on_equity_pct	>15% = efficient equity use; compare to sector median (Ch. 9)
Median P/E	Median pe_ratio (latest year)	Market willingness to pay per unit of earnings; high = expensive vs history
Median D/E	Median debt_to_equity	<0.5 = conservative; Financials excluded from interpretation
Total Companies	Count = 100	Confirms coverage; drops if the year filter has missing rows
Median Rev CAGR 5Y	Median revenue_cagr_5y_pct	Top-line compounding; >12% is strong for large caps
Debt-Free Companies	Count where debt_to_equity==0 or icr_label==Debt Free	Quality filter proxy — investable without leverage risk

Tip: Debt-Free is defined strictly as D/E == 0 or icr_label == “Debt Free”. A company with D/E 0.02 still shows leverage — use the Screener D/E slider for finer control.

Sector Donut

Below the tiles, a Plotly pie with hole=0.55 shows the 10-sector distribution (percent + label). Hover reveals exact count and share. The donut reflects the static companies table and is not year-dependent.

Top 5 by Composite Quality Score

A table lists the 5 highest-ranked companies for the selected year, sorted by composite score — a z-normalized blend of ROE, ROCE, NPM, low D/E (inverted), FCF conversion and revenue CAGR. Higher is better. Copy a ticker into Company Profile or Trend Analysis to investigate.

Interpretation example: if the Top-5 are dominated by IT and Consumer Staples with ROE >20% and D/E <0.3, quality is concentrated in asset-light sectors — a useful prior before running the Screener Quality preset.

CHAPTER 5

Company Profile — Deep Dive

Open Company Profile (pages/02_profile.py). The screen is built for one ticker at a time and loads in under 0.5s (cached for 600s).

Search & Company Card

Search company by name or ticker — type to filter, then select from the dropdown. The search box runs live autocomplete over company_name and company_id (e.g. typing “REL” surfaces RELIANCE). If no match, a yellow banner reads “Ticker not found — please try another”. Once selected, a card shows name, sector, sub-sector, tickers and the about description.

KPI Tiles (Latest Year)

KPI	Source column	Healthy range (large-cap)
ROE %	return_on_equity_pct	>15% strong; >20% exceptional
ROCE %	return_on_capital_employed_pct	>14% indicates efficient capital use
Net Profit Margin %	net_profit_margin_pct	>10% for non-Financials; varies by sector
D/E	debt_to_equity	<0.5 conservative; 0 = Debt Free
Revenue CAGR 5Y	revenue_cagr_5y_pct	>10–12% sustained growth
FCF (Cr)	free_cash_flow_cr	Positive and growing; compare to Net Profit

Values show N/A when the latest year is missing. Interest Coverage is displayed as “Debt Free” when icr_label == Debt Free even if the numeric column is null.

10-Year Grouped Bar: Revenue & Net Profit

A grouped bar overlays Revenue (Sales) and Net Profit across up to 10 years. Use it to check whether profit is growing with sales (operating leverage) or diverging (margin compression). A blue info banner notes available history when <10 years.

ROE & ROCE Dual-Axis Line

Two lines share the X axis with left axis = ROE, right axis = ROCE. ROCE consistently below ROE suggests high leverage flattering ROE; ROCE above ROE is rare and often signals negative equity.

Pros & Cons — Badge Pills

Auto-generated pros (green) and cons (red) appear as pill badges (from src/nlp/pros_cons_generator.py + the prosandcons table). Examples: “High ROE 5Y avg”, “Debt Free”, “Declining margins”. Treat these as starting points for diligence, not recommendations.

Analyst Workflow for a Single Name

1. Search ticker → read the card and About text.
2. Scan the KPI tiles vs sector median (Chapter 9).
3. Check Revenue vs Profit bars — is growth profitable?
4. Check the ROE/ROCE spread — is leverage driving returns?
5. Read pros/cons pills, then validate each against the charts.
6. Jump to Trends (overlay 3 metrics) or Peers (radar) for context.

CHAPTER 6

Screener — Finding Opportunities

Open Screener (pages/03_screener.py) — the most-used screen for idea generation. All controls live in the sidebar so the results table uses full width.

The 10 Metric Sliders (Sidebar)

Slider	Type	Range	What it filters
ROE min	Minimum	0–30%	Return on equity floor
D/E max	Maximum	0–3.0	Leverage ceiling (Financials: interpret cautiously)
FCF min (Cr)	Minimum	0–5000	Free cash flow floor
Revenue CAGR min	Minimum	0–25%	5-year sales compounding floor
PAT CAGR min	Minimum	0–25%	5-year profit compounding floor
OPM min	Minimum	0–40%	Operating margin floor
P/E max	Maximum	5–60	Valuation ceiling
P/B max	Maximum	0.5–10	Price-to-book ceiling
Dividend Yield min	Minimum	0–6%	Yield floor
ICR min	Minimum	0–20	Interest coverage floor

All sliders operate on the latest year per company. Moving any slider re-filters instantly; the label above the table reads e.g. “23 companies match your filters”. Extreme values correctly show 0 results with no crash.

Six Preset Buttons

Preset	Intent	Typical slider fill
Quality	High profitability, low leverage, strong cash	ROE ≥15, D/E ≤0.5, OPM ≥15, ICR ≥5, FCF >0
Value	Cheap vs earnings/book	P/E ≤18, P/B ≤2.5, Div Yield ≥1%
Growth	Fast top & bottom line	Rev CAGR ≥12, PAT CAGR ≥12, ROE ≥12
Dividend	Income focus	Div Yield ≥2%, payout consistency, D/E ≤1
Debt-Free	Zero leverage	D/E = 0 (or icr_label Debt Free)
Turnaround	Improving but still cheap	Low P/E, rising OPM, moderate D/E

Presets are defined in config/screener_config.yaml and applied in src/screener/engine.py. Clicking a preset overwrites slider positions; you can then drag individual sliders.

Live Results Table & CSV Download

The table shows company_id, name, sector, composite score and all filtered metrics, sorted by composite score descending. Click **Download CSV** (st.download_button) to export the filtered view — well-formed even for 0 or 100 rows.

Screener Tips

- Start with a preset, then tighten one slider at a time — the count drop shows which constraint is binding.
- For Financials, relax D/E and focus on ROE/ROCE spread and ICR instead.
- Combine Quality + Value manually: set Quality, then cap P/E ≤20.
- Always check the Capital Allocation Map (Ch. 9) for a filtered name — a cheap P/E with an Aggressive Growth pattern may be burning cash.

CHAPTER 7

Peer Comparison

Open Peer Comparison (pages/04_peers.py). The goal is to answer: is this company better or worse than its natural peers on each metric?

Peer Group Dropdown

A dropdown lists all 11 peer groups from the peer_groups table (56 memberships). If peer_groups is empty, the screen falls back to companies.broad_sector.

Radar Chart — Company vs Peer Median

Select a ticker within the group. The radar (Plotly Scatterpolar, fill=toself) plots 8 metrics normalized 0–1 across the peer set: ROE, ROCE, NPM, OPM, Revenue CAGR 5Y, PAT CAGR 5Y, P/E (inverted), D/E (inverted). Normalization is min-max per metric within the group; D/E is inverted so larger = better. The blue polygon is the selected company; the orange polygon is the peer median.

Reading the radar: a “star” shape (all axes extended) = all-round quality. A spike on leverage but a dent on growth = conservative balance sheet, weak top-line.

Side-by-Side KPI Table

Below the radar, a table lists every company in the group with the same 8 KPIs plus composite score. The benchmark row (peer median) is highlighted. Sort by any column to find the group leader on ROE, lowest D/E, etc.

When Peers Mislead

- Groups are based on sector, not size — cross-check with the Sector Analysis bubble where bubble size = Market Cap.
- Financials vs non-Financials: D/E and ICR medians are not comparable.
- Missing data: a peer with N/A on a metric is excluded from that axis median.

CHAPTER 8

Trend Analysis

Open Trend Analysis (pages/05_trends.py) — the time-series complement to Company Profile. Instead of tiles for one year, you see trajectories for one company across up to 10 years.

Company Search & Metric Selector

Company — selectbox over the universe with label “TICKER — Company Name”. Metrics — multi-select, overlay up to 3 at once (Sales, Net Profit, ROE, ROCE, NPM, OPM, D/E, P/E, FCF and CAGR variants). Selecting more than 3 is blocked.

10-Year Line Chart with YoY Annotations

A Plotly line chart with markers plots each selected metric over year. Each point carries a YoY % annotation: +x% in green for positive, -x% in red for negative. The formula is (current - prior)/|prior|*100; N/A breaks the line (gap) rather than interpolating.

Data Table with YoY %

Below the chart, a table mirrors the plotted data — one row per year, one column per metric, plus YoY %. Copy it into a note; it is the audit trail for any YoY claim.

Interpretation Framework

Pattern	What it suggests	Next check
Sales up, NPM flat, ROE up	Leverage-driven returns — check D/E trend	Peers radar: D/E axis
Sales up, NPM up, ROE up	Operating leverage — healthy compounding	Screener Growth preset
Sales up, FCF down	Growth funded by capex/WC — check cashflow	Capital Map pattern
Volatile YoY (±20%)	Cyclical or one-off items — normalize	Sector median bar (Ch. 9)

CHAPTER 9

Sector Analysis & Capital Allocation Map

This chapter covers two complementary screens often used together: Sector Analysis (06_sectors.py) for cross-sectional positioning and Capital Allocation Map (07_capital.py) for cash-flow behaviour.

Sector Analysis — Bubble Chart

Sector dropdown → bubble chart: X = Revenue (Sales), Y = ROE %, bubble size = Market Cap, colour = sub_sector. Each bubble is one company; hover shows ticker, revenue, ROE and market cap.

How to read it: top-right = large, profitable leaders; top-left = small but highly profitable (potential quality mid-caps); bottom-right = large but low ROE (turnaround or capital-heavy). Colour clusters reveal sub-sector effects.

Sector Median KPI Bar Chart

Below the bubble, a bar chart shows the sector median for 5 KPIs: ROE, ROCE, NPM, D/E, Revenue CAGR 5Y (median across companies in the sector, latest year), followed by a full median table. Use medians, not means, to avoid skew from outliers.

Capital Allocation Map — Treemap

A treemap groups all companies by 8 capital allocation patterns derived purely from the signs of CFO, CFI, CFF for the latest year:

Pattern	CFO	CFI	CFF	Interpretation
Healthy Growth	+	–	–	Generates cash, invests, returns/deleverages
Aggressive Growth	+	–	+	Generates cash but funds growth with financing
Mature Returner	+	–/+	–	Cash cow returning capital
Turnaround Investing	–	–	+	Burning operating cash, needs funding
Restructuring	–	+	–/+	Selling assets to fund operations
Cash Builder	+	+	–	Harvesting — low reinvestment
Distressed	–	–	–	All outflows — stress signal
Financial Engineering	–/+	+/–	+	Financing-dependent — check notes

The treemap is sized by |CFO| (larger cash generators dominate visually), coloured by sector, with drill-down on click. A pattern selector below lists companies in the chosen pattern sorted by CFO descending.

Using Them Together

Workflow: 1) In Sector Analysis, spot a low-ROE large bubble. 2) Check its sector median bar — is low ROE sector-wide or idiosyncratic? 3) In Capital Map, find its pattern — if “Distressed” or “Turnaround Investing” with negative CFO, the low ROE is cash-backed and warrants caution. Conversely, a small top-left bubble with “Healthy Growth” is a classic quality-screen hit.

CHAPTER 10

Annual Reports & Tearsheet PDFs

Open Annual Reports (pages/08_reports.py). Every company-year with a filing is listed with a clickable BSE PDF link from documents.report_url (1,584 rows).

Company Search & Year Listing

The screen lists available annual-report years for a selected company, most recent first, each as a hyperlink (SELECT * FROM documents WHERE company_id=? ORDER BY year DESC).

404 Handling & Offline Grace

Before rendering a link as “live”, the app attempts a HEAD request with a short timeout:

- **200 OK** → normal link.
- **404** → red badge “Report unavailable” next to the link (still shown).
- **Offline / timeout** → grey note “(availability not checked — offline)” and the link renders without a check. The screen never blocks on network.

Tearsheet PDF — One Click per Company

A download button offers reports/tearsheets/{ticker}_tearsheet.pdf when it exists. Tearsheets are 2-page, navy-header PDFs generated by src/reports/tearsheet.py (ReportLab + Matplotlib). There are currently 92 tearsheets.

- **Page 1 — Financial Overview:** header with name/ticker/sector; About paragraph; KPI tiles (ROE, ROCE, NPM, D/E, Interest Coverage, FCF Conversion); Sales & Net Profit bar; ROE/ROCE trend line.
- **Page 2 — Capital & Market:** Free Cash Flow bar; stock-price line; 10-year ratio table; investment-thesis footer. Each PDF is guaranteed ≥30 KB and uses matplotlib.use(Agg) for headless rendering.

Generating / Regenerating Reports

Task	Command	Output
Single company	python -c "from src.reports.tearsheet import generate_tearsheet; generate_tearsheet('RELIANCE')"	reports/tearsheets/RELIANCE_tearsheet.pdf
All classified companies	python -m src.reports.tearsheet — or — make report	92 PDFs
Sector reports	python -m src.reports.sector_report	10 sector-level PDFs

When to regenerate: after make load or make ratios, or when companies.about is patched. The batch generator skips failures with a printed warning, so a single bad ticker never aborts the run.

Research Tips

- Always open the latest 2 annual reports for a new name — check auditor qualifications and related-party notes before trusting the ratios.
- If a BSE link is 404, search the ticker on bseindia.com manually; the red badge is a hint, not a verdict.
- Attach the tearsheet PDF to any screening CSV when sharing — it gives the recipient the 10-year context behind the screen hit.

CHAPTER 11

Valuation Module & Interpretation Guide

The valuation layer is intentionally simple and transparent — it lives in src/analytics/valuation.py and prefers the real market_cap source (with ev_ebitda) over synthetic fallbacks. Run python -m src.analytics.valuation to regenerate outputs.

What It Computes (Latest Year, 100 Companies)

Metric	Formula	Source
FCF Yield %	FCF / market_cap_crore × 100	financial_ratios.free_cash_flow_cr / stock_prices.market_cap_crore
Sector Median P/E	Median pe_ratio per broad_sector	financial_ratios.pe_ratio grouped by companies.broad_sector
P/E vs Sector Median %	(company P/E – sector median)/sector median ×100	Derived
Flag	Caution / Discount / Fair	Rule-based

Flag rules: Caution if P/E > sector_median × 1.5; Discount if P/E < sector_median × 0.7; otherwise Fair. Null market_cap_crore or pe_ratio is median-filled — every row gets a flag, never N/A.

Outputs

File	Rows	Key columns
output/valuation_summary.xlsx	100 + sector_median sheet	company_id, company_name, sector, P/E, P/B, EV/EBITDA, FCF_yield_pct, 5yr_median_PE, PE_vs_sector_median_pct, flag
output/valuation_flags.csv	Caution + Discount only	Same subset with supporting data for quick review

Ratio Glossary — What Each KPI Means

Ratio	Definition	Healthy / note
ROE %	Net Profit / Avg Equity ×100	>15% strong; high ROE with high D/E = leverage, not efficiency
ROCE %	EBIT / Capital Employed ×100	>14% efficient; compare to the ROE spread
Net Profit Margin %	Net Profit / Sales ×100	>10% (non-Financials); sector-dependent
D/E	Total Debt / Equity	<0.5 conservative; Financials structurally higher
Interest Coverage (ICR)	EBIT / Interest Expense	>3 comfortable; Debt Free if no debt
FCF Conversion %	FCF / Net Profit ×100	~80–120% healthy; persistently <60% = WC/capex strain
Revenue CAGR 5Y	(Rev_latest/Rev_5y_ago)^(1/5)–1	>10% sustained growth for large caps
P/E, P/B, EV/EBITDA	Price / Earnings, Book, EV / EBITDA	Lower = cheaper; always vs sector median
Dividend Yield %	DPS / Price ×100	>1.5% meaningful for large caps
OPM %	Operating Profit / Sales ×100	Core operating efficiency; compare to the NPM spread

Best Practices & Pitfalls

Do:

- Always compare a ratio to its sector median (Chapter 9) before calling it high/low.
- Use FCF Yield alongside P/E — a low P/E with negative FCF yield is a value trap.
- Check the Capital Allocation pattern for any Discount flag — “Distressed” + Discount = cheap for a reason.
- Re-run make ratios and python -m src.analytics.valuation after any ETL change.

Don't:

- Don't screen Financials on D/E alone — use ICR and the ROE/ROCE spread instead.
- Don't extrapolate a 2-year CAGR — the platform shows 5Y CAGR to avoid recency bias.
- Don't treat SIMULATED stock_prices as live — valuation flags are indicative.
- Don't ignore N/A — a missing P/E shown as Fair is a data gap, not a verdict.

Quick Reference — Commands

Task	Command
Load warehouse	make load — or — python -m src.etl.loader
Compute ratios	make ratios
Run valuation	python -m src.analytics.valuation
Launch dashboard	streamlit run src/dashboard/app.py --server.port 8501
Launch API	uvicorn src.api.main:app --port 8000
Run tests	pytest tests/ -v
Generate tearsheets	python -m src.reports.tearsheet

APPENDIX A

Acceptance Checklist Cross-Reference

This appendix maps the deliverables verified in docs/acceptance_checklist.pdf to the guide chapters where each is explained. Evidence below is read live from the warehouse and filesystem when this guide is generated.

ID	Deliverable	Path	Guide Ch.	Live evidence
D-01	nifty100.db	data/nifty100.db	Ch 2,3	2,708 KB, FK check 0
D-02	load_audit.csv	output/load_audit.csv	Ch 2	companies 100 / PL 1263 / BS 1137 / CF 1152 / FR 0 / SP 5520
D-03	validation_failures.csv	output/validation_failures.csv	Ch 2	329 rows, 0 CRITICAL
D-04	exploratory_queries.sql	notebooks/exploratory_queries.sql	Ch 2	1 KB
D-05	financial_ratios table	data/nifty100.db → financial_ratios	Ch 2,5,11	1,263 rows, 45 cols, peer_percentiles 7,150
D-06	capital_allocation.csv	output/capital_allocation.csv	Ch 9	73 KB, 1,152 rows
D-07	screener_output.xlsx	output/screener_output.xlsx	Ch 6	44 KB, 7 sheets
D-08	screener_config.yaml	config/screener_config.yaml	Ch 6	6 presets
D-09	peer_comparison.xlsx	output/peer_comparison.xlsx	Ch 7	29 KB, 11 groups
D-10	Radar charts	reports/radar_charts/	Ch 7	92 PNGs
D-11	Streamlit Dashboard	src/dashboard/app.py	Ch 3,4–10	8 pages, wide layout
D-12	valuation_summary.xlsx	output/valuation_summary.xlsx	Ch 11	100 rows, 2 sheets
D-13	cashflow_intelligence.xlsx	output/cashflow_intelligence.xlsx	Ch 9,10	170 KB
D-14	pros_cons_generated.csv	output/pros_cons_generated.csv	Ch 5	49 KB, 455 rows
D-15	analysis_parsed.csv	output/analysis_parsed.csv	Ch 2	4 KB
D-16	Company Tearsheets	reports/tearsheets/	Ch 10	92 PDFs (~2pg)
D-17	Sector Reports	reports/sector/	Ch 9,10	10 PDFs
D-18	Portfolio Summary PDF	reports/portfolio/	Ch 10	1 PDF
D-19	cluster_labels.csv	output/cluster_labels.csv	Ch 6,9	92 rows, 5 clusters
D-20	FastAPI Server	src/api/main.py	Ch 3	routers mounted
D-21	pytest_report.html	reports/pytest_report.html	Ch 3	161 KB, 147 tests
D-22	analyst_guide.pdf	docs/analyst_guide.pdf	All Ch	50 KB (this file, regenerated)
D-23	acceptance_checklist.pdf	docs/acceptance_checklist.pdf	App A	5 KB

Cluster segmentation (92 rows): Distressed or Turnaround 42 / Defensive Dividend Payers 18 / Emerging Growth 18 / Value Cyclicals 12 / High-Quality Compounders 2.

How to verify: run python -m src.etl.loader; python -m src.analytics.ratios; python -m src.analytics.cagr; python -m src.analytics.cashflow_kpis; python -m src.analytics.peer; python -m src.analytics.valuation, then check output/load_audit.csv counts and PRAGMA foreign_key_check=0. The checklist PDF is regenerated live from the filesystem at build time — no manual entry.

See docs/excel_integration.md for ETL details. This guide is generated from the codebase by generate_analyst_guide.py — if the dashboard and guide disagree, the dashboard is authoritative.